

lodge contaminants into satellites, or short circuit electrical components.

Other elements of a satellite that need testing during its construction, design, and payload include:

- Solar panels
- Antennas
- Batteries
- Electrical circuits
- Centre of gravity and mass measurements
- Information transmitting systems from satellite to the ground station
- Fuel cells

Currently, NASA and other private companies are trying to build smaller vehicles that would be capable of repairing satellites. NASA is also working on ideas that would help assemble satellites in space itself.

The satellite's assembly team is the first group to test it. Next, a quality control team conducts a separate series of tests. The team often repeats many of these tests to gather enough data to decide whether or not the satellite is fit to go into orbit.

Testing has to be carried out in a clean room. Scientists and engineers involved in testing can be usually seen wearing a white suit. A clean room or environment is an integral part of satellite testing because one tiny outside contaminant can produce a drastic effect on a satellite.

The two major satellite tests are as follows:

Antenna deployment test. The uplink/downlink commands and telemetry data are established once the satellite antenna is perfectly deployed in space. So, for establishing reliable communication between the ground control station and satellite, this test is essential.

Subsystem testing. It involves activities such as battery voltage and bus voltage testing, primary and secondary battery voltage testing, solar panel voltage testing, actuator and sensor testing, momentum wheel testing, magnetic voltage testing, sun sensor testing, etc.

Testing story of GOES-S

National Ocean and Atmospheric Administration tested its Geostationary Operational Environmental Satellite-S (GOES-S), a weather satellite, in



The United Launch Alliance Atlas V rocket stands ready for liftoff with NOAA's GOES-S satellite on Space Launch Complex 41 at Cape Canaveral Air Force Station in Florida (Credit: NASA)

March 2017. As a part of testing, the team had placed GOES-S in a thermal vacuum chamber to determine its ability to operate in the extreme cold conditions of space.

The vacuum chamber tested the satellite across four different cycles that ranged from intense cold to intense heat. Severe temperature fluctuations in the airless chamber gave scientists a chance to check how well the satellite's sensitive instruments perform in these harsh conditions. Scientists had also tested GOES-S thrusters, which helped orient its orbit after its launch vehicle released it.

During testing, a satellite's exact centre of gravity and mass are also measured. This ensures that the satellite is compatible with its launch vehicle. It also helps control the orientation when the satellite is in orbit. This even helps to know how long it will operate in space.

Has the time come for fully automated testing of satellites?

Let's take the example of Electrical Ground Support Equipment (EGSE) satellite testing solution developed by Atos, a French-based MNC. It makes sure that satellites are ready to launch and work perfectly. The Radio Check-Out Equipment (RF SCOE) is the industry-standard solution to test radio frequency subsystems of a satellite with the highest degree of precision. The payload EGSE is highly flexible as it tests the instruments of navigation, earth observation, and

telecom satellites. Finally, the power SCOE ensures that the satellite's vital power supply works reliably under all space conditions.

The RF SCOE performs fully automated S, X, Ku, and Ka-band measurements in the Telemetry, Tracking and Command (TT&C) section of the satellite during assembly of spacecraft and space environment testing at the test centre facility. It does the same in the L, S, Ku, Ka, and UHF bands for the mission communication and payload section.

The payload EGSE allows the fully automated execution of calibrations and tests, and it easily integrates with heterogeneous components from different providers and suppliers. Further, all test sequences and procedures are defined in the open source language TCL and can therefore be easily tailored and extended to the best-fit requirements of the satellite owner.

Key parameters seem to be on the rise

Thousands of satellites will be launched over the next decade to provide global Internet coverage, mobile telecommunications, earth observation analytics, and space-based Internet of Things (IoT) for real-time monitoring of assets. Testing the characters of microwave performance of antennas of such satellites is a major part of testing. Measuring group delays with unknown or unstable local oscillators and TVAC drifts are also an important